# MILITARY SPECIFICATION

## CONTAINER, TORPEDO MK 197 MOD 2

This specification has been approved by the Naval Ordnance Systems Command, Department of the Navy.

## 1. SCOPE

1.1 This specification covers Containers for Torpedoes MK 43 Mod 1, MK 43 Mod 3, MK 44 Mod 0, MK 44 Mod 1, MK 46 Mod 0 and MK 46 Mod 1. These containers are, for test purposes, of one type and one class and are referred to herein as the Container. (See 6.2 for description).

## 2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids form a part of this specification, to the extent specified herein.

SPECIFICATIONS

MILITARY

| | |
|---|---|
| WR-11 | Design and Test of Packaging, Packing, Shipping and Handling Equipment for Weapon System Components |

STANDARDS

MILITARY

| | |
|---|---|
| MIL-STD-129 | Marking for Shipment and Storage |

DRAWINGS

BUREAU OF NAVAL WEAPONS

BUWEPS LIST OF DRAWINGS

| | |
|---|---|
| LD 269273 | Container, Torpedo, MK 197 Mod 1 and all documents listed thereon |

FSC 1355

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| | |
|---|---|
| LD 269294 | Container, Torpedo, Mark 197 Mod 2 and all documents listed thereon |
| BUWEPS DWG 1803640 | Torpedo MK 43 Mod 3, Outline Drawing |
| BUWEPS DWG 2159700 | Torpedo MK 46 Mod 0, Outline Drawing |
| BUWEPS DWG 2539302 | Torpedo MK 46 Mod 1, Outline Drawing |

(Copies of specifications, standards, drawings, and publications required by the supplier in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction sample. Unless otherwise specified in the contract or order, preproduction samples of the Container shall be manufactured using the methods and procedures proposed for the production. The sample will be tested as specified in Section 4 herein and is for the purpose of determining that, prior to starting production, the contractor's production methods are capable of yielding items that comply with the technical requirements of the contract.

3.2 General requirements.

3.2.1 Materials and treatments of materials and parts. Unless otherwise specified, all materials, treatments, and parts used in the manufacture of the Container shall be in accordance with the requirements specified herein and with the applicable documents listed in LD 269273 for Mod 1 and LD 269294 for Mod 2.

3.2.2 Gages. The supplier shall provide those gages necessary and adequate to assure that the Container will meet dimensions and requirements shown on the applicable documents listed in LD 269273 for Mod 1 and LD 269294 for Mod 2. When requested by the supplier, the procuring agency may furnish existing, pertinent drawings of Naval final inspection gages for use only in the designing of the supplier's gages. This action does not relieve the supplier of the responsibility for designing and manufacturing the gages needed for the fulfillment of the contract.

3.2.3 Dummy load. For test purposes a dummy load simulating a torpedo shall be used. Configuration and weight distribution of the dummy load shall be in accordance with Drawing 2539302 for Container MK 197 Mod 1, and Drawing 1803640 for Container MK 197 Mod 2.

3.3 Performance requirement.

3.3.1 Airtightness. The Container shall show no evidence of air leakage when pressurized to 1.0 plus 0.5 minus 0.0 psig.



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3.4 Environmental requirements.

3.4.1 Vibration. The Container, in its normal shipping attitude using a dummy load, shall withstand vibration through the following frequencies and thereafter show no damage to the stored contents and meet the performance requirement of this specification.

3.4.1.1 Variable frequency. The Container shall withstand vibration through a frequency range sweep of 4 to 60 cps at 1 inch double amplitude or 1.3 g whichever is less.

3.4.1.2 Natural frequency. The Container shall withstand vibration at the natural frequency of the packaged-article cushioning system at 1.3 g. or 1 inch double amplitude, whichever is less, for a period of not less than 15 minutes in each of the three principal axis. When resonance is in the range of 9 to 20 cps, vibrating frequency shall be less than 9 cps with an input amplitude sufficient to produce 1.3 g's.

3.4.2 Rough handling. The Container, using a dummy load, shall withstand the following shock loading and thereafter show no damage to the stored contents and meet the performance requirement of this specification.

3.4.2.1 Rotational drop. The Container shall withstand shock loading as induced by a rotational drop of not less than 18 inches on a steel or concrete surface on each of the 4 base corners.

3.4.2.2 Impact. The Container, at a velocity of 8 feet per second, shall withstand impact on all four vertical faces against a rugged unyielding surface.

3.4.2.3 Tip-over. The Container shall withstand shock caused when it is tipped slowly on a steel or concrete surface from its base to each side.

3.5 Workmanship. Prior to and after final assembly, all components and parts of the Container shall be thoroughly cleaned of excess metal chips, flux, and other foreign matter. All parts and assemblies of the Container shall be constructed in a manner to assure compliance with this specification. Particular attention shall be paid to neatness and thoroughness of welding, markings of parts and assemblies, coatings and finishes and freedom of parts from burrs and sharp edges. The standards of workmanship exhibited in any approved preproduction sample, subject to any qualification stated in the Government notice of approval, shall be determinative of the requirements of the contract relative to workmanship insofar as not specifically covered by applicable specifications.



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## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. Inspection of the Container shall be classified as follows:

a. Preproduction and periodic production inspection (Paragraph 4.4)
b. Acceptance inspection (paragraph 4.5)

4.3 Sampling

4.3.1 Preproduction sample. A preproduction sample of one Container manufactured in accordance with 3.1 shall be subjected to the preproduction tests detailed in 4.4 at an activity designated by the procuring activity. Acceptance of the preproduction sample shall be based on no defects in the sample. Further production of the Container by the supplier, prior to the approval of the preproduction sample shall be at the supplier's risk.

4.3.2 Acceptance inspection sampling. Acceptance inspection shall be performed on each Container being procured.

4.3.3 Periodic production sampling. Periodic production samples shall be selected from units that have satisfactorily passed the acceptance inspection of 4.5.2. These samples shall be subjected to the tests of 4.4 at an activity designated by the procuring activity. Periodic production sampling shall be as follows:

Step 1. Upon initiation of production randomly select one (1) unit from one group of 50 units. Upon satisfactory compliance of the sample selected from this group, shift to Step 2.

Step 2. Randomly select one (1) unit from one group of 100 units. Upon satisfactory compliance of the sample selected from this group, shift to Step 3.

Step 3. Randomly select one (1) unit from each group of 200 units. Continue with Step 3 for the balance of the contract or until a defective unit is found.



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When a defective unit is found in any group, acceptance of all units on hand shall be withheld and the contractor shall:

a. Determine the cause of the defect.

b. Take the necessary actions to correct or eliminate the defect from all units on hand and, if so directed by the procuring activity, from those units previously shipped.

c. Submit to the procuring activity for approval, the proposed corrective action to prevent recurrence of the defect.

Return to Step 1 and proceed as indicated when inspection is resumed.

4.4 Preproduction and periodic production inspection. The preproduction and periodic production sample, after satisfactorily passing the acceptance inspections detailed in 4.5, shall be subjected to the following tests and examinations.

4.4.1 Test equipment

4.4.1.1 Vibration test machine. Vibration tests shall be conducted by any vibration machine capable of meeting the conditions specified in 4.4.2.1. Means shall be provided for controlling the direction of vibration of the machine and for adjusting and measuring the frequencies and amplitudes of vibration.

4.4.1.2 Hoist. A hoist capable of safely lifting a load of 1000 pounds.

4.4.2 Test procedures. The Container, using a dummy load, shall be subjected to the following tests at ambient temperature only:

4.4.2.1 Vibration. The loaded container shall be rigidly secured to the vibration equipment, in its normal shipping attitude, and vibrated in directions parallel to each of the three principal orthogonal axes, with frequency and applied force varying uniformly with time, in scans as shown in Table I.



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TABLE I

| Test No. | Frequency of the forcing function at limits of scan (cycles/sec.) | | Max. Accel or Amplitude Applied to Exterior of Pack | Test Duration |
|---|---|---|---|---|
| 1. | 4 | 60 | 1.3 g's or 1 inch double amplitude (whichever is less) | Sufficient to determine resonant point |
| 2. | *Natural frequency of packaged article - cushioning system | | same as above | 15 min. |

*When resonance is in the range of 9 to 20 cps, vibrate at a frequency below 9 cps with an input amplitude sufficient to produce 1.3 g's.

4.4.2.2 Rough handling

4.4.2.2.1 Rotational drop. The Container shall be subjected to the rotational drop test as specified in MR-11.

4.4.2.2.2 Impact. The Container shall be subjected to the impact test as specified in MR-11 except that the impact velocity shall be eight (8) feet per second.

4.4.2.2.3 Tip-over. The Container shall be subjected to the tip-over test as specified in MR-11.

4.4.2.3 Acceptance criteria. The Container shall be examined immediately after the vibration and rough-handling tests to determine the ability of the Container to protect contents. There shall be no evidence of a substantial amount of shifting of contents within the Container that would create conditions likely to cause damage during shipment. There shall be no evidence of binding as the contents are removed from the Container.

4.4.2.3.1 The Container shall be resealed and subjected to the acceptance test specified in 4.5.

4.5 Acceptance inspection.

4.5.1 Test equipment.

4.5.1.1 Compressor or pressurized tank. The compressor or pressurized tank shall be capable of pressurizing the Container to 1.0 plus 0.5 minus 0.0 psig. The compressor or pressurized tank shall have a pressure gage capable of indicating 3.0 psig and shall have a hose with a standard inflating chuck.



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4.5.2 Test procedure. The sealed container shall be pressurized to one (1) plus 1/2 minus 0 psig, using the compressor or pressurized tank described in 4.5.1.1, through the tank valve in the forward upper end cap.

4.5.2.1 A heavy soapy solution shall be applied with a brush to all seam welds and to the gasket sealed areas of the container.

4.5.3 Acceptance criteria. There shall be no bubble or other visible evidence of leakage.

4.6 Packing, labeling and marking. Packing, labeling and marking of the Container shall conform to Section 5 of this specification.

## 5. PREPARATION FOR DELIVERY

### 5.1 Packing

5.1.1 Level C. Unless otherwise specified, packing for domestic shipment shall be in accordance with approved commercial packing methods, adequate to protect the Container from damage and to assure delivery at destination. The method used shall be commensurate with shipping at the lowest practical commercial rates.

5.2 Labeling and marking. Unless otherwise specified, labeling and marking of all shipments shall be in accordance with Standard MIL-STD-129. Any special marking required by the contract or order shall be provided.

## 6. NOTES

### 6.1 Intended use.

6.1.1 Container Mark 197 Mod 1 is intended for use in the foreign and domestic shipment and storage of one of the following items:

(a) Torpedo Mk 44 Mod 0 - Warshot Assembly

(b) Torpedo Mk 44 Mod 1 - Warshot Assembly

(c) Torpedo Mk 44 Mod 0 - Main Assembly and Nose Section

(d) Torpedo Mk 44 Mod 1 - Main Assembly and Nose Section

(e) Torpedo Mk 46 Mod 0 - Warshot Assembly

(f) Torpedo Mk 46 Mod 1 - Warshot Assembly

(g) Torpedo Mk 46 Mod 1 - Main Assembly



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6.1.2 Container Mark 197 Mod 2 is intended for use in foreign and domestic shipment and storage of one of the following items:

(a) Torpedo Mk 43 Mod 1 - Warshot Assembly

(b) Torpedo Mk 43 Mod 3 - Warshot Assembly

6.2 Description. The Container is a fabricated metal cylinder, 113-9/16 inches long by 19 inches in diameter, divided longitudinally approximately in half so that it is top-opening. The upper shell weldment secures to the lower shell weldment by means of swing-bolts and flange-nut assemblies. Fork-lift pockets, skids and sling-hook rings are provided to facilitate handling. The lower shell weldment is provided with shock mounts. Strap assemblies, fastened to the shock mounts, are used to secure the torpedo in the Container. A humidity indicator is installed in the forward end of the upper shell weldment. The empty Container weighs approximately 300 pounds.

6.3 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Facility assigned to perform preproduction and product verification tests.

(c) Selection of applicable level of packing.

(d) Any exception to this specification.

(e) Responsibility for additional samples required because of failure of any unit(s) to pass the prescribed periodic production inspections.



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